M.A.R.I.A.
Stabilization and Full-Scale Capability Roadmap

Pre-effector hardening plan before unrestricted ClawBot access

Document type	Architecture and execution roadmap
Audience	Engineer or coding team responsible for system evolution
Scope	Long-term, non-MVP stabilization and capability sequencing
Context date	27 March 2026

Positioning. This roadmap assumes M.A.R.I.A. is being developed as a durable cognitive system with AGI-oriented architectural discipline. It explicitly avoids MVP shortcuts. The goal is not to make the system appear autonomous quickly, but to make it genuinely stable, auditable, memory-coherent, and safe under increasing authority.

1. Executive summary

• M.A.R.I.A. already has a rare advantage: the cognitive core is modularized, contracts exist, tests are extensive, and system-level functions such as homeostasis, sandbox learning, semantic memory, self-analysis, and creative tension handling are already present.

• The main risk is no longer “can the system do interesting things,” but rather “can all layers remain coherent as authority, memory volume, autonomy, and effector access increase.”

• The immediate objective is to harden the architecture before unrestricted ClawBot control: one truth pipeline for decisions, end-to-end observability, scheduler governance, bounded autonomy, and promotion discipline between sandbox and production.

• This document sequences the work in dependency order so the system grows upward from stable foundations instead of expanding sideways into hidden complexity.

2. Development stance

• No MVP framing. Every phase should move the system toward a durable operating architecture rather than a demo-friendly shortcut.

• Safety is not only about preventing harmful actions; it is about preventing silent degradation, hidden drift, unverifiable state, and self-optimization against local metrics.

• Effector authority must grow only when the cognitive substrate becomes inspectable and mechanically trustworthy under load, retries, and conflicting goals.

• External tools, models, and effectors are organs of the system. They must be orchestrated by governance rules, not by ad hoc convenience.

3. North-star architecture goal

Target state. A long-running cognitive system that maintains identity continuity, memory coherence, transparent decision traces, bounded experimentation, and progressive authority over external effectors without sacrificing auditability or architectural clarity.

Capability axis	Desired end-state
Memory	Semantic retrieval, beliefs, conversation memory, and episodic consolidation resolve into a consistent and freshness-aware knowledge surface.
Planning	Planner decisions are traceable to goals, inputs, confidence, provenance, and policy constraints.

Capability axis	Desired end-state
Autonomy	Self-analysis and experiments can propose and validate improvements without silently mutating production behavior.
Execution	Effectors operate through staged authority levels, explicit budgets, and action validation loops.
Governance	Every meaningful system change can be inspected historically: who proposed it, what evidence supported it, how it was tested, and why it was promoted.

4. Stabilization priorities in plain language

• First stabilize what the system knows.

• Then stabilize why the system chose a given action.

• Then stabilize how models and resources are allocated.

• Then stabilize how the system changes itself.

• Only then widen ClawBot authority.

5. Dependency map

Layer	Depends on	Why it must come first
Observability and decision traceability	Existing contracts, logs, planner hooks, homeostasis ticks	Without visibility, every later autonomy feature becomes harder to debug and easier to misinterpret.
Memory consistency layer	Observability, semantic memory, world model, persistence contracts	The system must know which knowledge surface is authoritative, fresh, and allowed to influence decisions.
Scheduler and resource governance	Observability, model registry, homeostasis metrics	Multi-model systems fail silently when routing and cold-start behavior are opaque.

Layer	Depends on	Why it must come first
Autonomy governance	Memory consistency, traceability, scheduler predictability	Self-analysis and experiments must operate on reliable state, not fragmented state.
Effector safety envelope	Autonomy governance, action safety, policy gates	External authority should only expand after internal discipline is enforceable.
Full ClawBot access	All previous layers	Unrestricted effector control is a consequence of proven stability, not a feature to test first.

6. Ordered execution plan

Phase 1 — Unified observability and decision traces

Objective: make every critical decision inspectable.

• Add a normalized decision trace object shared by planner, world model queries, policy checks, scheduler calls, effect validation, and effector commands.

• Record for each decision: input context, selected goal, supporting memories or beliefs, confidence, freshness, model/tool chosen, policy result, and outcome.

• Expose traces in Web UI and Telegram summaries in a compact but operator-readable form.

• Introduce correlation IDs spanning a single cognitive episode across perception, planning, model use, and execution.

Exit criteria. Exit criteria: an operator can reconstruct why any major action was taken without reading raw logs.

Phase 2 — Memory consistency and truth hierarchy

Objective: remove ambiguity between semantic retrieval, beliefs, episodic memory, and conversation-derived state.

• Define a formal truth hierarchy: raw source records, derived beliefs, semantic vectors, episodic summaries, operator instructions.

• Attach freshness, provenance, confidence, and invalidation metadata to all retrievable memory surfaces.

- Build a reconciliation layer so planner queries receive a single merged answer structure instead of independent fragments from multiple stores.

- Implement periodic re-indexing and invalidation rules for beliefs, hints, and derived memories after exams, corrections, or operator overrides.

Exit criteria. Exit criteria: planner and self-analysis consume one coherent knowledge surface with explicit provenance.

Phase 3 — Model scheduler hardening and resource governance

Objective: make multi-model orchestration deterministic under pressure.

- For every model invocation, log route selection reason, expected role, cold/warm status, execution duration, memory cost, and fallback chain outcome.

- Separate strategic routing logic from emergency degradation logic so REDUCED and SURVIVAL modes are predictable.

- Introduce explicit execution budgets for time, retries, RAM pressure, and concurrent heavyweight calls.

- Create health-aware route policies: when to prefer rule-based flow, local models, cloud models, Codex CLI, or defer to human approval.

Exit criteria. Exit criteria: model routing behavior is explainable, reproducible, and bounded under load.

Phase 4 — Autonomy governance and self-modification discipline

Objective: let the system improve itself without silently redefining itself.

- Make proposal, experiment, recommendation, sandbox validation, and promotion separate first-class states with audit trails.

- Require every experiment to declare target metric, blast radius, rollback path, and promotion conditions before execution.

- Prevent local metric gaming by adding cross-metric validation: no parameter change promotes if it improves one metric while degrading memory coherence, stability, or action validity.

- Record who or what approved the promotion: operator, policy rule, or future bounded-autonomy delegate.

Exit criteria. Exit criteria: self-analysis can improve the system, but every change remains attributable and reversible.

Phase 5 — Effector safety envelope for ClawBot

Objective: constrain external authority before full access.

- Define staged effector authority levels: observe only, suggest only, operator-confirmed execution, bounded autonomous execution, unrestricted execution.
- Attach action classes to policy budgets: message sending, task triggering, file/system actions, network actions, restart actions, multi-step chains.
- Require effect validation after each non-trivial action: did the world change as expected, and did any side-effect exceed the declared scope?
- Create anti-cascade guards: command rate limiting, cooldowns after failed actions, dependency locks, and abnormal retry detection.

Exit criteria. Exit criteria: ClawBot access is policy-shaped, budgeted, observable, and interruptible.

Phase 6 — Full authority readiness review

Objective: decide whether unrestricted ClawBot access is architecturally justified.

- Run long-duration stability tests with memory growth, model cold starts, multiple proposed goals, and simulated operator interruptions.
- Run authority drills: malformed commands, ambiguous goals, stale memories, conflicting priorities, missing dependencies, and partial failures.
- Require a formal readiness review covering memory coherence, traceability completeness, rollback ability, scheduler behavior, and effector post-condition checks.
- Only grant unrestricted access if failures remain bounded, explainable, and recoverable without human forensics.

Exit criteria. Exit criteria: full ClawBot access is a justified consequence of evidence, not optimism.

7. Cross-cutting workstreams that should run in parallel

Workstream	Purpose	Run in which phases
UI for operator trust	Turn hidden state into usable operator oversight.	Phases 1–6
Schema discipline and ADR updates	Keep every new capability anchored in a documented contract.	Phases 1–6

Workstream	Purpose	Run in which phases
Load and soak testing	Measure degradation over time instead of only pass/fail correctness.	Phases 2–6
Failure simulation	Practice ambiguity, stale state, and resource pressure before real-world effectors rely on the system.	Phases 3–6
Security and privilege review	Ensure more authority does not implicitly widen attack surface or failure radius.	Phases 5–6

8. What should not be done yet

• Do not grant unrestricted ClawBot authority simply because the command interface already exists.

• Do not add more external tools until the scheduler can explain and bound model/tool selection under pressure.

• Do not let semantic memory, world model, and conversation memory evolve independently without a unifying truth hierarchy.

• Do not promote self-analysis recommendations directly into production without audit-grade promotion rules.

• Do not mistake activity, creativity, or successful demos for architectural readiness.

9. Recommended gate model before full ClawBot control

Gate	Required evidence	Decision
Gate A — Traceability	Planner and action traces are complete and operator-readable.	Permit suggest-only effector mode.
Gate B — Memory coherence	Merged truth layer resolves provenance and freshness conflicts.	Permit operator-confirmed execution.
Gate C — Scheduler discipline	Routing, budgets, and fallbacks are stable under	Permit bounded autonomous execution.

Gate	Required evidence	Decision
	load.	
Gate D — Safe self-modification	Experiments and recommendation promotions are auditable and reversible.	Permit broader task autonomy.
Gate E — Effector validation	Post-condition checks and anti-cascade guards hold under adverse scenarios.	Consider unrestricted ClawBot access.

10. Concrete build order for the coding team

• DecisionTrace schema and storage contract.

• Trace hooks in planner, policy, scheduler, action safety, and effector dispatch.

• Unified memory query contract with provenance, confidence, and freshness.

• Re-indexing and invalidation pipeline for beliefs, hints, and semantic vectors.

• Scheduler telemetry and execution-budget enforcement.

• Proposal/experiment/promotion lifecycle with rollback metadata.

• Effector authority-level framework and post-condition verification.

• Readiness test suite for long-run autonomy and ClawBot escalation.

11. Final recommendation

Recommendation. Treat the next stage not as feature expansion but as architectural hardening. M.A.R.I.A. already has enough cognitive surface area to become genuinely powerful. What it now needs is a stronger spine: unified truth handling, full decision visibility, deterministic resource governance, disciplined self-change, and staged effector authority. If these layers are completed in order, unrestricted ClawBot access can become an earned capability inside a system that is moving toward long-horizon cognitive operation rather than toward a brittle imitation of autonomy.

Appendix A — One-line dependency chain

Contracts and existing core → traceability → memory coherence → scheduler governance → autonomy governance → effector safety envelope → full ClawBot authority